Kick-off Diagnostic · L6-7 — Geometric Sequences and Series

Grade 11

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).
The one idea: A geometric sequence multiplies by the same common ratio r, so $a_n = a_1 r^{n-1}$; adding its first n terms gives $S_n = \dfrac{a_1(1-r^n)}{1-r}$ when $r \neq 1$.
The mistake most students make: Writing r^n instead of r^{n-1}: $a_6 = 6 \cdot 2^6 = 384$ instead of $6 \cdot 2^5 = 192$.
Readiness items 3, 4, 5, 7, 8 test: exponent rules, the common ratio from terms and from a graph, exponential versus linear growth, sigma notation, running totals.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All four slides, including the readiness check
B. I watched and read but skipped the check
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

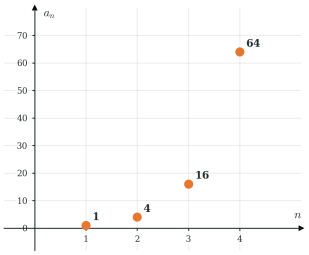
In the pre-session, which formula gives the nth term of a geometric sequence?

A. $a_n = a_1 + (n-1)r$
✓ **B.** $a_n = a_1 r^{n-1}$
C. $a_n = a_1 r^n$
D. $a_n = a_1 \cdot n \cdot r$

Signal. Wrong → did not study slide 2. A = the arithmetic formula; C = the warned r^n slip; D = multiplied by n and r.

3 FILL IN THE BLANK · 30 s — READINESS

The graph shows the first four terms of a geometric sequence. Its common ratio is $r =$ _____

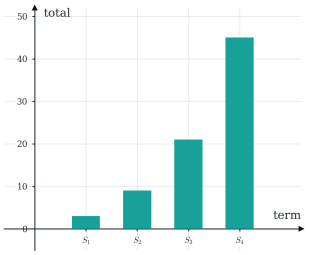


Accept: 4

Signal. 3 → subtracted $(4-1)$, a common difference; $\frac{1}{4}$ → divided the wrong way round. Finding r by dividing is not yet secure.

4 CHECKBOX · 30 s — READINESS

The bars show the running totals S_1, S_2, S_3, S_4 of $3 + 6 + 12 + 24$. Select the **two** true statements.



✓ **A.** $S_3 = 21$
B. $S_2 = 6$
✓ **C.** $S_4 = 45$
D. $S_4 = 24$

Signal. $S_2 = 6$ or $S_4 = 24$ → read a single term instead of a running total; a series is a sum, not a list.

5 MATCH · 30 s — READINESS

Match each sum with its value.

$\sum_{k=1}^{3} 2k$ → 12

$\sum_{k=1}^{3} k^2$ → 14 (shown shuffled in Wayground)

$\sum_{k=1}^{3} 2^{k-1}$ → 7

Signal. 6 → added $1+2+3$ and forgot to square; 36 → squared the total $(1+2+3)^2$; 14 for the last → used 2^k. Sigma notation needs a quick walk-through.

6 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

The pre-session warned about a_6 for 6, 12, 24, … Which wrong answer was it?

A. 36 B. 72

C. 64 ✓ **D. 384 (from $6 \cdot 2^6$)**

Signal. 36 = added 6 each time; 72 = multiplied by n; 64 = dropped the 6. Wrong → skimmed the common-mistake bar on slide 3.

7 REORDER · 30 s — READINESS

Put the steps for the sum of the first 5 terms of $2+6+18+\cdots$ in order.

Correct order: $a_1 = 2,\ r = \dfrac{6}{2} = 3 \rightarrow S_5 = \dfrac{2\left(1-3^5\right)}{1-3} \rightarrow S_5 = \dfrac{2(-242)}{-2} \rightarrow S_5 = 242$

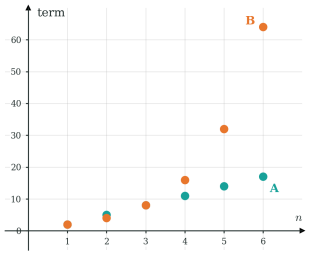
Signal. Wrong → naming a_1, r and n before substituting is the gap; expect the sum formula to be applied without r.

8 DROPDOWN · 30 s — READINESS

Sequence A adds the same amount each time; sequence B is geometric. Which statement is correct?

A. A has ratio $r = 3$ B. B has ratio $r = 4$

✓ **C. B has ratio $r = 2$** D. A has ratio $r = 2$



Signal. A $r = 3$ → took the common difference for a ratio; B $r = 4$ → divided 8 by 2, skipping a term; A $r = 2$ → read a_1 as the ratio.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach exponent rules and the common ratio with me. **3–4** → straight to the formulas; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.